

How to prove a polyhedron empty

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We consider a polyhedron P :

$$P = \{x \in \mathbb{R}^n : Ax \leq b\},$$

where A is an $m \times n$ matrix and b is an $m \times 1$ vector. The set $\text{rec}(P) := \{x \in \mathbb{R}^n : Ax \leq 0_m\}$ is called the *recession cone* of P .

When P is non-empty an easy argument exists to verify this, namely any $\bar{x} \in \mathbb{R}^n$ satisfying $A\bar{x} \leq b$. Specifically, one can compute $A\bar{x}$, and then compare $A\bar{x}$ and b . We say $\bar{x}$ is a *certificate* that proves P is non-empty. When P is empty, however, it is not that clear how to prove this. In this note we provide a certificate to prove P is empty.

The idea is to re-phrase the question in terms of valid inequalities. An inequality is *valid* for P iff the inequality holds for all points in P . One therefore has:

Observation 1 (*Equivalent condition for emptiness of P*)

The polyhedron P is empty $\iff$ The inequality $0_n^T x \leq -1$ is valid for P . ■

We can now use the following characterization of valid inequalities for P (proved in note 2.(F)):

Fact 1 (*Characterization of valid inequalities for a polyhedron*)

Let $\alpha^T x \leq \beta$ be an inequality, and suppose $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$.

$$\alpha^T x \leq \beta \text{ is valid for } P$$

$$\iff \alpha = A^T u \text{ and } \beta \geq b^T u \text{ for some } u \geq 0_m. \quad \blacksquare$$

The inequality $0_n^T x \leq 0$ is trivially valid for $\text{rec}(P)$. Hence Fact 1 can be applied with $\alpha = 0_n$ and $\beta = -1$, i.e., the inequality $0_n^T x \leq -1$ is valid for P if and only if there exists $u \geq 0_m$ satisfying $0_n = A^T u$ and $-1 \geq b^T u$.

One has $0_n = A^T u$ and $-1 \geq b^T u$ for some $u \geq 0_m$ if and only if there exists $u \geq 0_m$ satisfying $A^T u = 0_n$ and $b^T u < 0$. Combining this with Observation 1 gives the following short proof of emptiness of a polyhedron. The result is due to Farkas, and is therefore called "Farkas's lemma".

Lemma 1 (*Farkas's lemma*)

One of the following statements hold, but not both.

- (i) *The polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ is non-empty.*
- (ii) *The inequality $0_n^T x \leq -1$ is valid for P , i.e., there exists $u \geq 0_m$ that satisfies $A^T u = 0_n$ and $b^T u < 0$. ■*

Farkas's lemma gives the promised easy argument for proving P is empty:
A vector $\bar{u} \geq 0_m$ that satisfies $A^T \bar{u} = 0_n$ and $\bar{u}^T b < 0$ proves P is empty.